

Cindrapect Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-153/B

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The control strategy links critical quality attributes to critical process parameters identified during development. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

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impact to product quality. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Cleaning verification swab results were below the health-based exposure limits derived for the product. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

3.2.P.4 Control of Excipients

For the commercial formulation F-153/B, Tablettose 80 (Lactose monohydrate, diluent) is used.

The approach described herein is consistent with ICH Q8(R2) and FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

Formulation Development Rationale

The basis for selection of Tablettose 80 is robust tablet hardness across the compression force design space, demonstrated across three pilot-scale batches.

Registered Specification

The current approved commercial specification (SPEC-EX-4328 Rev 05) lists Lactose monohydrate as Tablettose 80.

Note: the superseded specification (pre-2019, withdrawn) listed FlowLac 100; it is retained here for historical traceability only and has no bearing on the current control strategy.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. No changes to the approved analytical procedures are proposed in this submission. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

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